

My Database Theory:

1. Difference between Data and Information.

Data is raw facts, figures or values that have not yet been processed or organised in a way that would have a meaningful impact.

Information is processes and organised data, that can be used for meaningful impact such as decision making.

The difference is that Data is the raw material that is required to be processed for it to become Information that can be used in a reliable way.

2. What is Metadata?

Metadata is also defined as 'data about data'. It is structured information, that has hidden details that describe something, such as a digital photo. The photo is the main part, but the metadata stores things like where the photo was taken and from what device. Metadata helps to describe or tell you about something.

3. What is DBMS and its advantages?

A **Database Management System (DBMS)** is a collection of programs that work to manage the database structure and to control data stored in said databases. It sits at the middle, in-between the user and the database receiving all requests and then translating them to fulfil requests.

Ultimately, a DBMS improves the efficiency and effectiveness of data management by providing:

- **Enhanced data access and security:** A DBMS manages databases and controls user access, allowing end-users to retrieve data efficiently while maintaining data privacy and enforcing security policies.
- **Better data integration:** By enabling multiple users to access and manage data in an organised way, a DBMS provides a more unified and comprehensive view of an organisation's operations.
- **Efficient data manipulation:** A DBMS allows users to quickly process and respond to ad hoc data requests, such as retrieving or updating information.

- **Higher-quality information:** By delivering accurate and timely information, a DBMS supports faster, more informed decision-making and improves end-user productivity.

4. *Operational Database vs Analytical Database*

Feature	Operational Database	Analytical Database
<i>Purpose</i>	Supports day to day operations of an organisation	Supports Analysis and decision making for the organisation
<i>Data Types</i>	Current and detailed data	Historical and aggregated data
<i>Usage</i>	Frequently used for retrievals and updates	Mostly used for reporting or complex or potentially historical
<i>Speeds</i>	Fast transactional processes	Fast Data analysis

Operational Database example:

Customer Relationship management (CRM) – They track daily customer interactions, sales leads and requests.

Analytical Database example:

Datawarehouse's – They support decision making processes and can run complex queries for historical data used as insights for organisations.

5. *Types of data and use cases where NoSQL databases are effective*

Non-relational databases (NoSQL) databases are a flexible approach to data storage. They are designed to handle huge amounts of unstructured, or semi-structured data.

Semi-structured data – they do not follow any strict formats but do contain markers, an example would be an XML file which could be comma delimited.

Unstructured data – This is data with no defined format such as photos, videos and social media.

Use cases for NoSQL databases would be:

- Social media platforms such as a platform storing blog posts where each post would be different
- Image finding search engines such as a search engine using an image based on visual features, rather than the keyword search.
- Ecommerce platforms using it to advise that ‘someone else bought this’ item when using suggestions.

6. *Which DMBS does not require server configuration and its advantages?*

SQLite is a DMBS that does not require a server configuration.

It is commonly used for Mobile applications and small-scale projects due to its simple and self-contained nature.

Its advantages are:

- Compact in size
- Reliable
- Speed
- Straight forward to use

7. *What are ACID properties in the context of DBMS.*

in most relational DBMSs there are the built-in mechanisms to ensure data integrity and consistency:

ACID stands for:

Atomicity - A transaction is treated as a single unit and cannot be divided into small units

Consistency - A transaction that refers to the correctness of the database.

Isolation - Transactions operate in parallel, without affecting with each other.

Durability - Once a transaction is committed, the data remains saved even if a system failure occurs.

ACID guarantees that database transactions are complete, accurate, independent, and permanent, even in the event of system failures.